

Evaluating 3D Visual Fatigue Induced by VR Headset Using EEG and Self-attention CNN

Haochen. Hu*

Beijing Institution of Technology

Yue Liu†

Beijing Institution of Technology

Kang Yue‡

Beijing Institution of Technology

ABSTRACT

3D visual fatigue is one of the major factors that hinder the development of virtual reality contents towards larger population. We proposed an EEG-based self-attention CNN model to evaluate user's 3D visual fatigue in an end-to-end fashion. We adopted a wavelet-based convolution to extract spatiotemporal information and prevent overfitting. Besides, a self-attention layer was added to the feature extractor backbone to cope with the subject-variation problem in EEG-decoding. The proposed method is compared with four state-of-the-art methods, and the results demonstrate that our model has the best performance among all methods in subject-dependent and cross-subject scenarios.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Social and professional topics—User characteristics—People with disabilities

1 INTRODUCTION

Stereoscopic display technology is widely used in head-mounted display devices. However, most users develop the symptoms of visual fatigue induced by watching three-dimensional content [2], which seriously affect the user's experience of VR equipment, and hinder the development of VR equipment in a broader market.

Traditional methods for evaluating visual fatigue include the utilization of optometric devices on visual system and EOG data. However, the former is time-consuming and the latter faces challenges in finding reliable statistical-correlated features. The use of non-invasive EEG data containing rich time-frequency domain information related to visual information processing makes it possible to evaluate visual fatigue more comprehensively, therefore, it is widely adopted by researchers [2].

In recent years, deep learning has shown a strong capability in the fields of computer vision and natural language processing, which provides the possibility of establishing end-to-end models for sequential signal classification. In this paper, a deep learning model was proposed to classify 3D visual fatigue datasets, and Morlet wavelet convolution was used as the backbone of feature extractor. The subject invariant problem of EEG signal was solved by dictionary learning using self-attention feature transformer which extracts context information of EEG signal. The implementation detail of the proposed model can be found at <https://github.com/brandonlist/fatig>.

2 RELATED WORKS

EEG information related to 3D visual fatigue is mainly contained in the frequency domain. In particular, the statistical relationship between power spectrum density (PSD) characteristics of each frequency band and 3D fatigue level has been widely studied. With

the increase of time for subjects to watch uncomfortable 3D content, PSD in beta band and theta band increased significantly. Literature [3] used RDS to induce 3D fatigue, and found that PSD in alpha band increased with the increase of fatigue degree.

Traditional machine learning methods are often preceded by using multiple frequency band filters and spatial filters. With the development of deep learning technology, convolutional neural network is used to classify EEG signals. Literature [6] studied the relationship between the number of layers and accuracy of CNN networks. Literature [1] adopted depth-wise convolution to improve accuracy by minimizing the parameter number of the model.

3 METHODS

3.1 Dataset description

In total, 22 healthy subjects from the University of Chinese Academy of Sciences participated in the experiment, who are aged from 22 to 25 years old (average age 23, 4 females), and all right-handed. Besides, all subjects have normal vision. Random-Dot Stereogram (RDS) with nine different disparities was used to induce 3D visual fatigue in a VR environment. The experiment was divided into 6 blocks, 4.5 Min. each. 9 kinds of disparities appeared randomly. After each block, a questionnaire was used to investigate the degree of visual fatigue ranging from 1-3 where 1 means no fatigue and 3 indicates extreme fatigue.

30 channel-EEG data was collected by using Compumedics neuroscan scan synaps 2 system referenced to left mastoids (A1) according to the international 10-20 system. The sampling rate is 1000Hz. Six subjects' data were removed due to poor signal quality. To remove artifacts, artifact subspace reconstruction (ASR) was applied for preprocessing. The data from 500ms to 1000ms before the start of visual stimulation were extracted for each subject, with a total of 1200 trials.

3.2 Self-attention MorletInceptionNet

3.2.1 Morlet-Inception Backbone

Deep learning model used to classify EEG data is at risk of overfitting with a high-dimensional feature space and a relatively small amount of data compared with computer vision applications, where massive data is required to achieve SOTA performance. In previous studies, shallow structures were adopted to minimize the parameters of the EEG classification model. For example, depth wise convolution is used in [1] to reduce the amount of parameters, and in [6], stacked one-dimensional convolution is adopted instead of two-dimensional convolution. In this paper, we used the structure proposed in [7] where Morlet wavelet function is used as convolution kernel to extract time-frequency features in EEG. The wavelet function is formulated as follows:

$$\omega(t) = e^{-\frac{a^2 t^2}{2}} \cos(2\pi b t) \quad (1)$$

Only two parameters, i.e., a and b control the shape of the convolution kernel, which makes it possible to use a small number of fixed parameters to deal with variant length of segmented features, so as to reduce the complexity of each layer of network for stacking. At the same time, EEG patterns induced by different stimuli have scale

*e-mail: haohao9806@163.com

†e-mail: liuyue@bit.edu.cn

‡e-mail: einhep@bit.edu.cn

Table 1: Mean accuracy of different methods on 3D Fatigue classification task using 5-fold cross validation. Deep4 and Shallow refer to Deep4Net and ShallowConvNet proposed in [6], while Morlet refers to MorletInceptionNet proposed in [7]

Subject	Deep4	Shallow	EEGNet	Morlet-SA	Morlet
s1	0.5572	0.6667	0.59165	0.7031	0.6229
s2	0.6718	0.7687	0.6031	0.7843	0.7385
s3	0.8562	0.875	0.8447	0.8791	0.8718
s4	0.4812	0.601	0.4781	0.6927	0.625
s5	0.5156	0.7583	0.6322	0.7833	0.7718
s6	0.8552	0.8833	0.8406	0.8812	0.8875
s7	0.6145	0.6635	0.5833	0.6979	0.65
s8	0.6177	0.7125	0.5947	0.677	0.6979
s9	0.6218	0.6885	0.6333	0.7322	0.6968
s10	0.551	0.6072	0.5218	0.6404	0.5885
s11	0.746	0.7539	0.6851	0.8289	0.8031
s12	0.8864	0.8937	0.8677	0.951	0.8864
s13	0.8145	0.8604	0.776	0.876	0.8687
s14	0.4989	0.6406	0.5697	0.7083	0.6781
s15	0.4429	0.6703	0.507	0.664	0.6609
s16	0.8046	0.7734	0.7281	0.8289	0.8023
AVG	0.6584	0.7385	0.6535	0.7705	0.7406
Cross Subjects	0.6612	0.7208	0.6774	0.7607	0.7401

differences in time and frequency dimensions, and the inception module is added to solve the scale problem of different levels of characteristics.

3.2.2 Dictionary-learning Using Self-attention

Due to the individual differences of EEG signals, the performance of the model trained on other subject's data will deteriorate on the target subject. Methods to solve this subject-variant problem include transforming features into a subject-invariant space, or explicitly modeling the change of decision function by re-weighting and fine-tuning. Dictionary learning is used in [4], making use of both methods to find a common basis for all subjects' spatial patterns in an unsupervised way. They apply L-1 constraint to find specific transformation for each subject and each session. At the inference stage, the features are adaptively converted to a specific subspace, which significantly improves the accuracy of the subsequent classifier.

Deep learning is widely used for EEG single-trial classification. To the author's best knowledge, few methods are proposed to solve the problem of individual differences using end-to-end structure. In recent years, self-attention mechanism has been widely used to strengthen contextual information of feature. In machine translation, for example, the state vector extracted by using RNN may lose context information due to its compression characteristics, while the attention mechanism uses all the information input into sentence instead of the last few inputs, thus significantly boosting classification performance.

The input information is expressed in the form of attention weight [5]. In the context of EEG decoding, we use the self-attention mechanism to transform the feature extracted by using Morlet-Inception backbone module to a higher dimensional feature space, i.e. the dictionary. Then L1 constraints are used to force the self-attention module to learn specific sparse solutions for each subject and each session, so as to improve the generalization performance of the subsequent model.

4 RESULTS

The proposed model outperforms all deep learning architectures in both subject-dependent and cross-subject scenarios, indicating the effectiveness of wavelet-based CNN to extract joint spatio-temporal features and to avoid over-fitting. To verify the effect of self-attention

mechanism on the improvement of the model on the cross-subject dataset, the performance differences between self-attention Morlet inception model and the original model are compared. The results show that the accuracy is improved by 2.05 % after the addition of the self-attention layer, indicating the effectiveness of the proposed structure to generalize over subjects.

We also observed in ablation study that the accuracy of the suboptimal convolution networks shallowconvnet, eegnet and deep4convnet is also improved after the addition of self-attention layer, but it is not as significant as Morlet Inception model, indicating that the self-attention mechanism has a selective effect on the optimization of backbone feature extraction.

5 CONCLUSION

In this paper we proposed a novel method to automatically detect the level of 3D visual fatigue by using EEG and deep learning technique. And a self-attention layer was adopted to enhance the cross-subject performance and prevent over-fitting. Results demonstrate that the proposed method outperforms all the state-of-the-art models, both in subject-dependent and cross-subject scenarios. Future work will focus on the visualization of the shifted feature space, thereby confirming the ability of the network to learn task-relevant and subject-invariant features.

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